D02 · Inventory Strategy & Optimization

04 · Inventory reduction programs & working capital

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1 WHAT IT IS

Inventory reduction programs lower working capital by attacking the drivers that require inventory — demand and supply variability, lead time, lot size, and forecast error — so the reduction holds without sacrificing service, instead of mandating a top-down cut to the inventory level itself, which lowers the symptom while the cause rebuilds it.

2 WHY IT MATTERS

Little 1961 **(Little, 1961 — peer-reviewed)** proved L = lambdaW: the average inventory in a system equals throughput multiplied by flow time. The relationship is structural — inventory is the consequence of how much flows and how long it takes. A program that cuts inventory (L) without reducing flow time (W) can only reduce throughput (lambda): the cut shows up as stockouts and lost service, not as efficiency. This is why mandated inventory targets so often fail twice — **service collapses, and the inventory returns within a cycle or two** because the variability and lead time that generated it were never touched. The level is the symptom; the drivers are the disease.

3 HOW IT IS DONE

1. **Decompose the inventory into its drivers.** Split the balance into cycle stock (lot size), safety stock (variability and service), in-transit (lead time), pre-build (capacity/seasonality), and excess/obsolete. You cannot reduce what you have not attributed; the decomposition tells you which lever to pull.
2. **Attack lead time to cut in-transit and safety stock.** Shorter, more reliable lead times reduce both pipeline inventory and the safety stock that covers lead-time uncertainty. Lead-time reduction is often the single highest-leverage move because it lowers two components at once.
3. **Reduce variability to cut safety stock.** Better forecasts, demand sensing, smaller and more frequent orders, and supplier reliability all shrink the variability safety stock exists to absorb. Cutting the variability lets the buffer fall without lowering service — the only sustainable way to reduce safety stock.
4. **Right-size lot sizes to cut cycle stock.** Where setup or order cost has fallen, smaller lots cut cycle stock; this ties directly to changeover reduction on the shop floor. Cycle stock is a deliberate trade-off, not a fixed cost — revisit it as setup economics change.
5. **Clear excess and obsolete as its own track.** SLOB reduction is a distinct lever from structural inventory and should be run as a parallel disposition program, not folded into the target as if it were healthy stock to squeeze.
6. **Set targets on drivers, and protect service.** Make the program's KPIs lead time, forecast error, and variability — with service as a guardrail that cannot be breached. A target on the inventory number alone invites the service-destroying cut; targets on drivers produce a reduction that holds.

WORKED EXAMPLE — ILLUSTRATIVE

A business mandated a 20% inventory cut by quarter-end — **baseline: cut hit the number, then service fell from 97% to 89% and inventory rebounded 16% within two quarters as planners re-buffered to protect service**. It restarted as a driver-based program: decomposed the balance, cut a key lead time from 6 weeks to 3, improved forecast accuracy on A/X items, and ran SLOB disposition separately. Results: **inventory down 18% and held, service back to 97%, working capital released without the rebound** — because the drivers, not the level, were cut.

IN PRACTICE

Decompose before you target

Start every reduction program by attributing the balance to its components. The decomposition reveals where the inventory actually is — often excess and in-transit, not safety stock — and prevents the reflexive mistake of squeezing the buffer that is protecting service.

Lead-time reduction as a first move

Prioritize cutting and stabilizing lead times — supplier, manufacturing, and transport. Because lead time drives two inventory components and improves responsiveness, it usually delivers the largest sustainable working-capital release per unit of effort, with service rising rather than falling.

Service as a guardrail, not a casualty

Run the program with service as a hard guardrail that cannot be breached for an inventory number. A reduction that costs service has not freed capital — it has deferred a cost to lost sales and expedites. The guardrail is what keeps a working-capital program from becoming a service-destruction program.

Target drivers, measure both

Set program KPIs on the drivers — lead time, forecast error, variability, lot size — and measure inventory and service together as outcomes. Driver targets produce durable reduction; outcome-only targets (the inventory number) produce the rebound. Measuring both keeps the program honest about whether the cut will hold.

WHAT YOU WOULD USE

WORKING-CAPITAL / INVENTORY-REDUCTION ANALYTICS

SAP IBP / o9 / Kinaxis — driver decomposition & optimization — Inventory decomposition, driver-based what-if, and optimization that ties lead-time and variability changes to the inventory and service outcome. **(SAP, o9, Kinaxis — vendor)** The choice when reduction must be modeled against its drivers, not mandated on the balance.

Blue Yonder / RELEX — inventory analytics — Strong inventory decomposition and reduction analytics with service trade-off visibility. **(Blue Yonder, RELEX — vendor)** A solid fit where the reduction program is anchored in replenishment and item-location policy.

Working-capital / inventory-diagnostic specialists — Dedicated diagnostic tools that attribute the balance to drivers and quantify reduction opportunity. **(specialized vendors — vendor)** The vertical option when the priority is a one-time diagnostic to find and size the levers.

For operations that don't need a platform — A decomposition of the balance into its components in a workbook, with lead-time and variability reduction tracked as the real KPIs and service as a guardrail. Little's Law is the whole insight — attack flow time and variability, and the level follows. No platform is required to apply an identity.

4 SO WHAT

Operations that reduce inventory by cutting lead time and variability release working capital that stays released; operations that mandate a cut to the number trade service for a temporary balance-sheet win, then watch the inventory rebuild. The mandated inventory target is the most common and most self-defeating reduction program — it treats the symptom and is structurally guaranteed by Little's Law to either lose service or rebound. Real reduction is a driver program wearing an inventory goal.

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